BRANDON RIVERA-MELO

Software Engineer | Healthcare AI and Integrations

briveramelo@gmail.com | 630 688 6402
brandon.riveramelo.com | linkedin.com/in/briveramelo | github.com/briveramelo

Hands-on CTO and software engineer with 9+ years building healthcare and consumer products. Connects conversational AI to EHRs and real clinical workflows, with ownership spanning software architecture, implementation, production operations, and HIPAA compliance. Brings biomedical engineering, psychology, and business training to practical problems in patient access.

EXPERIENCE

MedCalls AI

2025 - Present

Chief Technology Officer | Hands-on software engineering

Own the software that turns AI receptionist conversations into appointment bookings and actionable office workflows, integrating Retell voice agents with healthcare data and practice operations.

- **Enabled end-to-end appointment bookings.** Built systems connecting AI receptionist conversations to patient lookup, availability search, and confirmed appointment creation.
- **Made EHR scheduling work across vendors.** Built a shared MedCalls patient and appointment model with Go adapters mapping to AdvancedMD, Tebra, and NexHealth. Unified REST/MCP operations let agents find patients and book, reschedule, and cancel appointments while respecting vendor-specific data and scheduling rules.
- **Made practice configuration usable by agents.** Built EHR imports and caching that turn provider, location, service, and appointment-type data into configured appointment choices. Connected these workflows to Airtable and n8n so operations staff can maintain what each receptionist offers.
- **Led HIPAA compliance and technical implementation.** Coordinate with Compliancy Group and guide vendor and technology choices around BAA coverage and PHI handling. Translate requirements into access-controlled Google Cloud services, account permissions, audit logging, secret management, and documented data-flow boundaries.
- **Kept core lookups dependable.** Built validation, targeted caching, and operational diagnostics for patient search and appointment retrieval, supporting these time-sensitive steps in live calls.
- **Connected calls to staff follow-through.** Built post-call processing, voicemail rendering, notification workflows, and a React/TypeScript dashboard so staff can review calls, act on messages, manage support cases, and see booking outcomes and usage.
- **Developing more controllable AI conversations.** Building a Go LLM-backed runtime to replace Retell conversation flows, targeting better reliability and cost control. Explicit confirmation gates, structured outputs, session traces, and synthetic evaluations support verification before live rollout.

CORE TECHNOLOGIES

Go, Python, TypeScript, React, Fastify, FastAPI, REST, MCP, SQL, Firestore, BigQuery, Cloud Run, Pub/Sub, Cloud Storage, Firebase Auth, Terraform, Docker, Git, n8n, Airtable, Retell AI.

BRANDON RIVERA-MELO

ADDITIONAL EXPERIENCE

Digital Health, LLC | *Founder and Lead Engineer* Founded Jan 2024

- **The Pocket Optimist.** Built an LLM classification pipeline grounded in explanatory-style research to support well-being through guided feedback and training. Extracts and validates event/explanation pairs, rates belief patterns, and generates personalized reports for coaching realistic optimism.
- **Smart Chart.** Built an AI-enabled patient-management platform for small clinics, bringing billing and scheduling workflows together with Python, React, Firebase, and Google Cloud.

HealthTech Apps, Inc. | *Head of Engineering* Jun 2023 - Jan 2024

- **MindKloud Journey.** Led the move from an iOS-only app to a shared Unity/C# codebase, launching on both mobile app stores. Built AWS-backed multimedia journaling, transcription, mood history, and engagement analytics to help young adults reflect on emotional patterns.

University of Utah Dec 2019 - Dec 2023

Software Design Engineer and Project Manager

- **Tilt Tracker: real-time feedback.** Co-inventor and lead engineer of a wheelchair-mounted system supporting pressure-injury prevention; owned firmware, APIs, and web software. Converted sensor streams into pressure-relief detection and personalized audio, visual, and haptic feedback on timing, angle, and duration.
- **Tilt Tracker: clinical analytics.** Built longitudinal adherence reports and interactive event-level analytics so patients, families, and clinicians could review patterns against personalized goals. Supported clinical deployment and co-authored peer-reviewed research, including ACM IMWUT.
- **NRH Control.** Coordinated engineers, clinicians, therapists, IT, and artists to deliver accessible hospital-room controls through touch, voice, and sip-and-puff input. Used patient testing to guide tutorials and consistent interfaces.
- **Bobby the Bladder and AD with Sam.** Led production and clinical integration of interactive spinal-cord-injury education. Deployed Bobby the Bladder to every patient iPad at the rehabilitation hospital and delivered an immersive exhibit explaining autonomic dysreflexia.

Age of Learning | *Software Engineer II* May 2018 - Sep 2019

- **Cut ABCmouse startup time by 50%.** Worked with the engineering team to redesign asset loading and caching and fix memory leaks, addressing crashes on low-memory devices used in schools and libraries.
- **Enabled large-scale experimentation.** Integrated Leanplum analytics for A/B tests with thousands of users per group. Contributed to 25+ iOS and Android updates in roughly 18 months, including feature integration and Android 64-bit compatibility.

Hathos Interactive | *Lead Engineer* Jan 2017 - May 2018

- **Clawface: 70,000+ downloads.** Led engineering and co-developed a Steam Workshop level editor for players to create and share content. Maintained playable builds every two weeks through launch.

Independent | *Lead Developer* Sep 2016 - Nov 2017

- **Derpy Dinos: nearly 50,000 downloads.** Led engineering and production through Steam release, coordinating a cross-functional team and building Unity/C# gameplay with leaderboards and achievement tracking.

EDUCATION

University of Utah | MBA, Information Systems certificate, 2022
MEAE, Game Engineering and Production, 2018

Northwestern University | BS, Biomedical Engineering; second major in Psychology, 2014
Certificates in Human-Centered Design and Entrepreneurship